
The Convergence Gap: Instruction-Tuned Language Models Stabilize Later in the Forward Pass

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Final outputs hide when a checkpoint commits to its next-token prediction. We in-
2 troduce the convergence gap, a model-diffing diagnostic that decodes each layer's
3 next-token distribution and measures its distance to the model's own final distri-
4 bution. Across six paired pretrained and instruction-tuned checkpoints in native
5 prompting regimes, instruction-tuned checkpoints remain farther from their final
6 predictions later into the stack. The effect persists under endpoint-matched raw
7 and tuned readouts, endpoint-free same-history checks, and fixed-history template
8 replay. Matched-prefix interventions identify late MLP windows as the largest
9 tested leverage point: late IT grafts into PT hosts increase late KL by +0.34 nats,
10 while PT-late swaps into IT hosts reduce it by -0.51 nats; matched random late
11 perturbations give only +0.003 versus +0.327 for the true late graft. A preselected
12 Gemma case study provides behavior-facing plausibility for the same late swap,
13 without serving as a benchmark claim. These results identify a robust prediction-
14 dynamics signature of post-training: released instruction-following checkpoints
15 tend to settle later, and late MLP computation is the strongest tested bidirectional
16 handle on that delay under matched histories.

17 1 Introduction

18 Model diffing needs diagnostics that compare paired checkpoints without reducing the comparison
19 to final-output scores. A useful diagnostic should ask not only whether two checkpoints end at
20 different predictions, but when each checkpoint settles on its own prediction during the forward
21 pass. We introduce the convergence gap as such a diagnostic and use pretrained/instruction-tuned
22 checkpoint pairs as the main test case.

23 We turn that question into a layerwise metric. For each layer, we decode a next-token distribution
24 and compare it to the model's own final distribution. The resulting **convergence gap** asks how far
25 the current layer is from the prediction that the model will finally output. A larger late gap means
26 the model is still changing its mind later in the stack.

27 Across six released pretrained/instruction-tuned checkpoint pairs, instruction-tuned models show a
28 larger late convergence gap. This is not just the familiar observation that late layers sharpen logits.
29 We compare each model to its own endpoint, match endpoint confidence and entropy, and also
30 use endpoint-free same-history checks that do not compare to the final distribution at all. The gap
31 remains.

32 Result in one place.

- 33 • Instruction-tuned checkpoints show a larger late convergence gap across six PT/IT pairs.
- 34 • Endpoint matching preserves the gap: raw +0.425 nats and tuned +0.762 nats.
- 35 • Endpoint-free checks agree: adjacent JS +0.052 and future top-1 flips +0.203.

- Matched-prefix MLP interventions identify late windows as the largest tested leverage point: late IT graft +0.34 nats and late PT swap -0.51 nats.
- Matched random late perturbations do not reproduce the effect: +0.003 nats versus true +0.327.
- A preselected Gemma behavior-facing case study shows native IT is preferred over the late PT swap on assistant-register prompts in 94.7% of pairwise comparisons; this is a high-signal bridge, not a benchmark claim.

The paper makes three contributions.

1. **A prediction-dynamics target for post-training.** The convergence gap measures when a model settles toward its own final next-token prediction, making stabilization through depth a model-diffing object.
2. **Endpoint and history controls.** Raw/tuned endpoint-matched readouts and endpoint-free same-history statistics test whether the gap is only a confidence, probe, or endpoint artifact.
3. **Matched-prefix MLP leverage.** Symmetric graft/swap experiments and matched random late perturbations show that late MLP windows are the largest tested bidirectional handle on delayed stabilization.

A preselected Gemma case study then asks whether the same late-window handle can matter in generation. It is deliberately secondary: it checks behavior-facing plausibility for the clearest family, not whether the convergence gap predicts benchmark performance or instruction following across tasks.

Scope matters. We do not claim a full mechanism for instruction following, a deployment-level behavior estimate across tasks, or a universal steering direction. The paper is about layerwise prediction dynamics under matched histories in released PT/IT checkpoint pairs, with one behavior-facing Gemma case study.

This paper deliberately does not test whether a late-stack effect ports across upstream states from another checkpoint at a selected token. Its primary object is the layerwise stabilization trajectory itself.

2 Setup

2.1 Models and Histories

We compare six paired pretrained (PT) and instruction-tuned (IT) checkpoints: Gemma 3 4B, Llama 3.1 8B, Qwen 3 4B, Mistral 7B, OLMo 2 7B, and DeepSeek-V2-Lite. The main discovery curves use all six pairs. Matched-prefix causal averages are reported on the dense families; DeepSeek is treated as a side case because MoE routing changes the interpretation of swapping dense MLP windows.

All comparisons are made under matched token histories. The discovery curves aggregate per-token layer readouts; the raw/tuned token-step counts range from 1.66e5 to 1.38e6 per checkpoint branch, with exact counts in Appendix A. For graft/swap interventions, the token prefix is held fixed while one MLP window is replaced. Bootstrap intervals are over prompt clusters where prompt-level rows are available; endpoint-matched controls additionally report matched-token retention and post-match balance.

2.2 Convergence Gap

Let p_1 be the next-token distribution decoded from layer 1, and let p_L be the model’s own final next-token distribution. The layerwise convergence gap is:

$$KL(p_1 || p_L).$$

This is a within-model quantity. It does not ask whether PT and IT have the same final distribution. It asks how far each checkpoint is from its own final prediction at each layer.

We use two companion checks. **Endpoint-matched convergence gap** matches PT and IT token steps by endpoint confidence, entropy, and top1/top2 margin before comparing late gaps. **Endpoint-free same-history checks** compare PT and IT native same-layer output distributions under identical teacher-forced token histories, avoiding each model’s final endpoint.

2.3 Matched-Prefix MLP Interventions

Matched-prefix interventions use overlapping early, middle, and late MLP windows of roughly 40% model depth, centered at the corresponding depth region. For example, Llama 3.1 8B uses layers 0-12, 9-21, and 19-31; exact per-family ranges are in Appendix B. On the PT host, `B_early`, `B_mid`, and `B_late` replace a PT MLP window with the corresponding IT MLP window. On the IT host, `D_early`, `D_mid`, and `D_late` replace an IT MLP window with PT. This gives a symmetric bidirectional leverage test for delayed stabilization under matched histories.

A separate random-control follow-up replaces the true late IT-minus-PT residual effect with matched random residual-projection perturbations. This tests whether any matched late perturbation recreates the convergence-gap signature.

3 Results

3.1 Instruction-Tuned Models Stay Farther From Their Final Predictions

The tuned-lens discovery view shows a broad late convergence gap across all six families. IT curves remain higher later in the stack, meaning the decoded distributions are farther from the final distribution at the same relative depth.

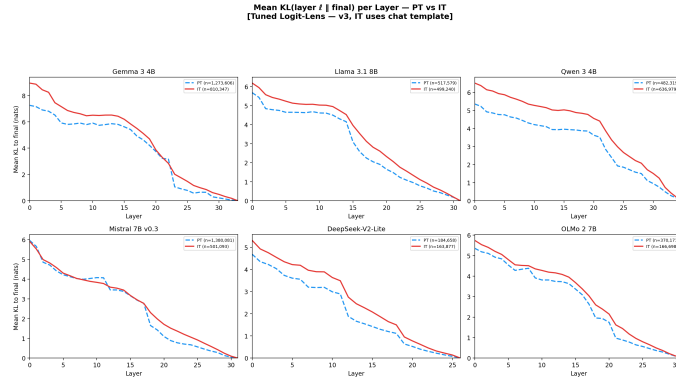


Figure 1: Mean KL(layer to final) per layer for six PT/IT families. IT curves remain higher through much of the stack, indicating a broader convergence gap.

Endpoint matching is the strongest control for this observation. We match token steps on final confidence, entropy, and top1/top2 margin, then remeasure the late gap. The IT-minus-PT gap remains positive in both raw and tuned readouts.

Check	Estimate	Interpretation
Endpoint-matched raw late KL	+0.425 nats [+0.356, +0.493]	raw-lens gap remains
Endpoint-matched tuned late KL	+0.762 nats [+0.709, +0.814]	tuned-lens gap remains
Endpoint-free adjacent JS	+0.052 JS [+0.048, +0.057]	IT has more remaining layer-to-layer movement
Endpoint-free future top-1 flips	+0.203 flips [+0.190, +0.215]	IT changes top-1 later

The matching quality is clean: minimum matched retention is 0.796, maximum post-match standardized mean difference is 0.057, and malformed rate is zero. The convergence gap is therefore not explained away by endpoint confidence, entropy, or a single tuned-lens threshold.

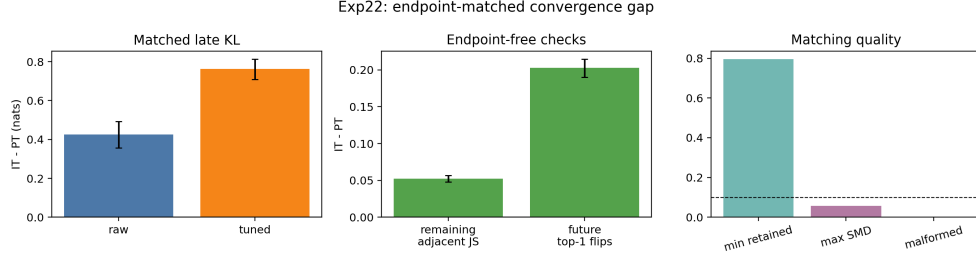


Figure 2: Endpoint-matched convergence-gap controls. Matching endpoint confidence and entropy does not remove the IT-minus-PT late gap, and endpoint-free stability checks remain positive.

106 Because instruction-tuned checkpoints are released for serialized chat interfaces, native chat templates are the main prompting condition rather than a cosmetic wrapper. Appendix A removes
 107 free-generation length confounding by generating one IT-native continuation per prompt and replaying
 108 the same forced token IDs through PT raw, IT native-chat, and IT raw/no-template cells. Under
 109 this fixed history, same-prompt late-KL effects are +1.181 nats for native-chat IT versus PT raw and
 110 +0.549 for raw/no-template IT versus PT raw; the native-minus-raw template delta is +0.632. Thus
 111 raw/no-template serialization attenuates the signature but does not erase it, and the native-template
 112 result survives forced continuation replay.
 113

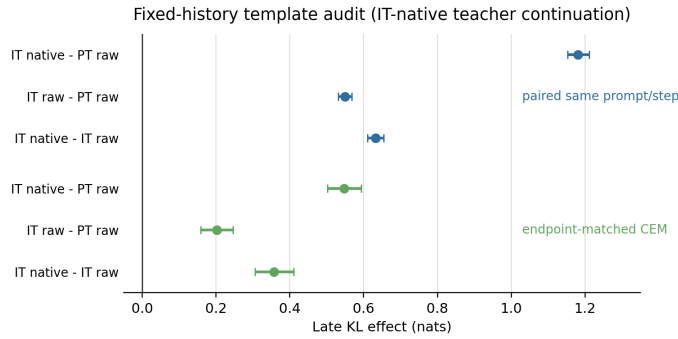


Figure 3: Fixed-history template audit. The native-chat effect survives forced continuation replay; raw/no-template replay remains positive but attenuated.

114 3.2 Late MLP Windows Have the Largest Tested Bidirectional Leverage

115 Native curves establish the phenomenon; matched-prefix interventions test where the delayed-
 116 stabilization signature is easiest to move. On the PT host, replacing late PT MLPs with IT MLPs
 117 recreates the delay much more than early or middle grafts. In the dense-family mean, final-20% KL
 118 changes by +0.34 nats for the late graft, compared with -0.03 early and -0.05 middle.

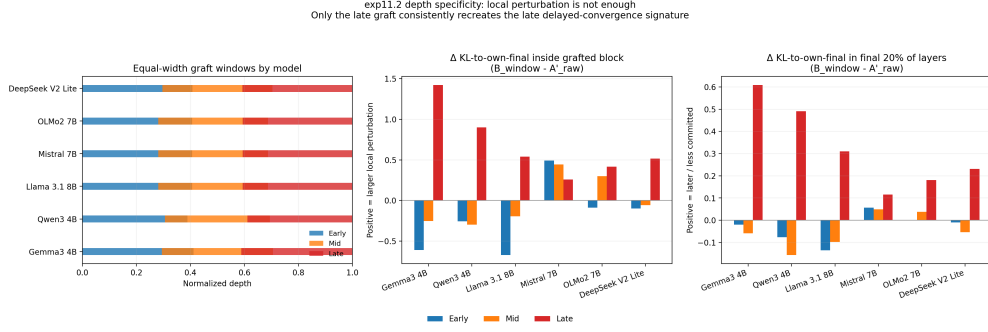


Figure 4: Depth-specific matched-prefix IT-MLP grafts into PT hosts. Late grafts most strongly recreate the delayed-stabilization signature.

119 The mirrored IT-host swap points to the same window. Replacing IT late MLPs with PT late MLPs
 120 reduces the IT delay by -0.51 nats, compared with -0.10 early and -0.23 middle. The same late
 121 window is therefore the largest tested bidirectional MLP handle on the convergence-gap metric.

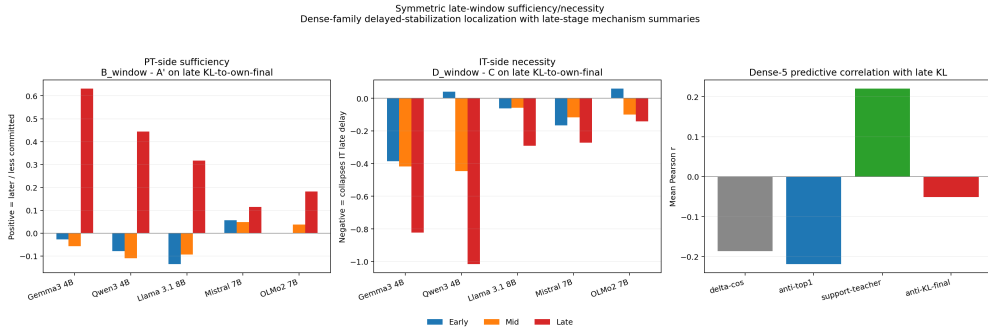


Figure 5: Symmetric matched-prefix graft/swap leverage. Late MLPs are the largest tested bidirectional window for delayed stabilization.

Intervention	Early	Middle	Late
IT MLP graft into PT host	-0.03	-0.05	+0.34
PT MLP swap into IT host	-0.10	-0.23	-0.51

122 The sign is not the important part for both rows; the matched direction is. IT late grafts increase
 123 the IT-like delay in PT hosts, while PT late swaps reduce the IT delay in IT hosts. Both directions
 124 identify late MLPs as the strongest tested window for moving the stabilization trajectory. Figure
 125 5 also makes the family heterogeneity visible: Gemma and Qwen are largest, but all five dense
 126 families have positive late-graft and negative late-swap point estimates.

127 Appendix B stress-tests this interpretation with a late-window width/center audit. The swap side
 128 remains stable at the late-full final-20% window (-0.625 nats, CI excluding zero), while the down-
 129 stream final-20% late-full graft is smaller (+0.070, CI crossing zero) and the pre-late swap control
 130 also moves. The edited late-window graft effect is clearer (+0.365, CI excluding zero). We therefore
 131 read the intervention evidence as late-window leverage on the convergence-gap metric, not as proof
 132 of a sharp late-only module or a complete circuit.

133 3.3 Matched Random Late Perturbations Do Not Reproduce the Effect

134 Late layers are close to the unembedding, so a skeptical explanation is that any sufficiently large late
 135 perturbation changes convergence. We test that directly with matched random residual-projection
 136 controls. The Exp19 control uses 120 prompts per dense family and three late random seeds; the

137 true late graft changes final-20% KL by +0.327 nats, while the matched random late perturbation
 138 gives only +0.003 nats.

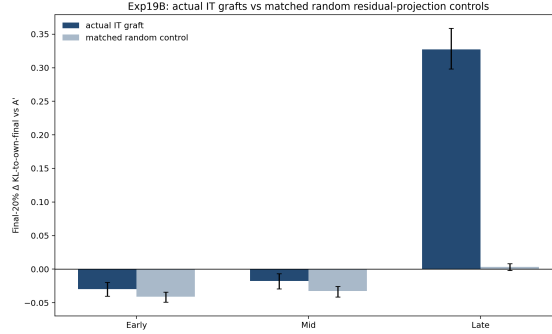


Figure 6: True late grafts versus matched random late residual-projection controls. Random perturbations do not reproduce the delayed-stabilization signature.

139 This does not prove that the real update is uniquely efficient or that every late random baseline is
 140 exhausted. It rules out the simplest magnitude-only account within this control family: the observed
 141 delayed-stabilization effect is not reproduced by this matched random late residual-projection per-
 142 turbation family.

143 3.4 Gemma Behavioral Case Study

144 The convergence gap is not itself a behavior metric. To check whether the same late-window handle
 145 can matter under natural decoding, we add an illustrative Gemma case study using the matched-
 146 prefix intervention family. Gemma is selected before this paper-facing analysis because it has the
 147 largest dense-family bidirectional late-window effects in Appendix B (+0.609 late graft, -0.822 late
 148 swap). This is a case study, not a cross-family behavioral claim.

149 Under free-running 512-token generation on the Exp15 600-prompt support, we compare native
 150 Gemma IT (`C_it_chat`) to the same IT host with a late PT MLP swap (`D_late_ptswap`). Blind
 151 pairwise judging prefers native IT over the late PT swap on assistant-register prompts in 94.7%
 152 of cases [91.6%, 97.3%] and on safety/format prompts in 93.3% [86.7%, 98.7%]. The pointwise
 153 assistant-register score drops from 4.733 to 3.929, while the early PT swap leaves that score essen-
 154 tially unchanged (4.751). Programmatic content checks do not show a matching collapse: MMLU
 155 forced-choice accuracy changes from 0.517 to 0.483, and reasoning exact match changes from 0.850
 156 to 0.925.

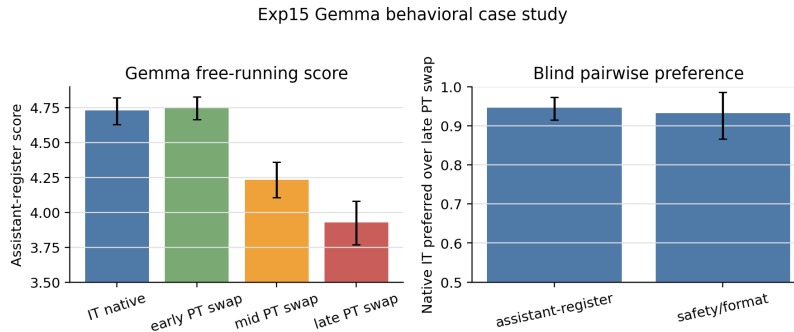


Figure 7: Gemma behavioral case study. Native IT is strongly preferred over a late PT MLP swap under free-running generation, while early swaps preserve the assistant-register score.

157 This result is best read as a behavior-facing bridge for the clearest family. It tests whether the
 158 late-window intervention that moves the convergence-gap metric can also matter downstream un-

der generation, while the load-bearing claim remains the layerwise stabilization result. It does not establish that the convergence gap predicts broad instruction-following behavior across families or benchmarks.

4 Discussion

4.1 Prediction Dynamics Rather Than Final Scores

The convergence gap reframes post-training as a change in the route to a prediction, not only the endpoint. The key observation is not that late layers sharpen logits; it is that released instruction-following checkpoints remain farther from their own final distribution later in the stack, even after endpoint matching and under endpoint-free same-history checks.

4.2 What the MLP Interventions Show

The matched-prefix graft/swap results are targeted counterfactual tests on a stabilization metric. They isolate which tested MLP window has the most leverage on the delayed-stabilization signature under matched histories, without requiring a full account of how the host model builds the upstream state. This is the right interpretation of the causal evidence: late MLP windows are the largest tested handle on the effect, but the result is still distribution- and metric-specific.

Just as important is what the intervention does not show. It does not say that late MLPs are a self-contained instruction-following module, that earlier layers are irrelevant, or that the late window is a sharp architectural boundary. A later-stabilizing trajectory can still depend on upstream representations; this paper isolates a prediction-dynamics effect rather than reconstructing the full circuit that produces it.

4.3 Relation to Prior Work

The convergence gap builds on lens-based prediction tracing, but it uses that tracing as a paired post-training diagnostic. Prior layerwise work asks what latent predictions look like or how earlier and later logits differ; here the comparison is between matched PT/IT checkpoint pairs and each model’s own endpoint. The diagnostic is useful because it turns “when does the model settle?” into a reproducible quantity that can be endpoint-matched, replayed under fixed histories, and targeted by interventions.

Line of work	Main target	What is added here
Tuned/logit lenses	decode latent predictions through depth	paired PT/IT convergence to each model’s own final distribution
Layer-contrast decoding and stages	exploit or describe differences between earlier and later layers	post-training diagnostic for when predictions stabilize
Feed-forward vocabulary updates	interpret MLPs as additive prediction updates	matched-prefix MLP graft/swap leverage on stabilization
Instruction-tuning model diffs	study behavioral and representational shifts after instruction tuning	depthwise prediction-stabilization signature across released PT/IT pairs
Activation patching and causal tracing	localize components for chosen output metrics	counterfactual tests on the model’s stabilization trajectory

The layer location also fits recent instruction-tuning representation studies. Zhao, Ziser, and Cohen divide Llama 2 SFT into shared early layers, transition layers, and refinement layers, with their refinement range covering layers 16-32; our dense-family late windows such as Llama 19-31, Mistral 19-31, OLMo 19-31, Gemma 20-33, and Qwen 22-35 overlap that refinement regime. The overlap is suggestive rather than definitive: Zhao et al. study task-specific representation similarity, while this paper measures next-token prediction stabilization and interventions on matched histories.

Feature-level model-diffing work is the natural next rung down in granularity. Crosscoders learn shared sparse dictionaries across layers or models, and recent chat-tuning crosscoder work argues that some chat-specific latents are interpretable and causally effective while also warning about sparsity artifacts. The convergence gap is coarser than such feature-level analyses, but it provides a diagnostic target: if a future crosscoder or SAE explains the delayed-stabilization signature, it would connect the layerwise phenomenon here to named latents and downstream behaviors.

198 This positioning keeps the claim scoped. The convergence gap is not a direct estimate of behavior,
199 and the graft/swap results do not by themselves identify a named sparse feature or full circuit. They
200 show a robust diagnostic signature and the largest tested MLP handle for moving that signature
201 under the evaluated distributions.

202 4.4 Limitations

203 The main limitations are probe dependence, intervention scope, template regime, and model-family
204 scope. Tuned-lens curves are estimators, so we pair them with raw-lens and endpoint-free checks.
205 Grafts and swaps are constructed matched-prefix interventions, not natural deployment trajectories.

206 For instruction-tuned checkpoints, chat templates are part of the released trained interface: they
207 serialize the conversation format and control tokens under which the models are normally optimized
208 and used. Appendix A reports dense-family fixed-history audits: native-template replay preserves
209 the late gap, raw/no-template replay remains positive but attenuated, and a PT-teacher reverse replay
210 has the same sign but weaker template separation with imperfect endpoint balance. We therefore
211 treat no-template prompting as an out-of-distribution serialization that modulates expression of the
212 gap, not as a replacement for the trained IT interface.

213 Appendix B adds a late-window width/center audit: the late-full swap effect is stable, but the graft-
214 side final-window estimate is small and the pre-late swap control also moves. We therefore claim late
215 MLPs are the strongest tested bidirectional handle, not a sharp layer boundary or a self-contained
216 module. DeepSeek-V2-Lite is included in discovery curves but not pooled into dense-family causal
217 averages because MoE routing changes the intervention interpretation. Finally, the paper does not
218 identify a named sparse feature circuit; it identifies a robust layerwise signature, the largest tested
219 MLP window for moving that signature, and one behavior-facing case study. A natural next step
220 is broader behavioral validation: testing whether late-gap magnitude predicts instruction following,
221 refusal, format adherence, calibration, or prompt sensitivity across families.

222 5 Conclusion

223 Instruction-tuned checkpoints do not merely end at different next-token distributions; they tend
224 to settle toward those distributions later in the forward pass. The convergence gap makes that
225 prediction-dynamics change measurable. Across six PT/IT pairs, the gap survives endpoint match-
226 ing, endpoint-free same-history controls, and fixed-history template replay. Matched-prefix grafts
227 and swaps identify late MLP windows as the largest tested leverage point on the delayed-stabilization
228 signature, matched random late perturbations do not reproduce it, and a preselected Gemma case
229 study shows behavior-facing consequences for the same late PT swap.

230 The resulting claim is compact and distinct: **instruction-tuned checkpoints in this sample stabilize**
231 **later, and late MLPs are the strongest tested bidirectional handle on that delay.**

References

- Belrose, N., et al. (2023). Eliciting Latent Predictions from Transformers with the Tuned Lens. arXiv:2303.08112.
- Chuang, Y., et al. (2024). DoLA: Decoding by Contrasting Layers Improves Factuality. *ICLR 2024*.
- Du, H., et al. (2025). How Post-Training Reshapes LLMs: A Mechanistic View on Knowledge, Truthfulness, Refusal, and Confidence. *COLM 2025*.
- Geva, M., Schuster, R., Berant, J., and Levy, O. (2022). Transformer Feed-Forward Layers Are Key-Value Memories. *EMNLP 2022*.
- Geva, M., Caciularu, A., Wang, K. R., and Goldberg, Y. (2022). Transformer Feed-Forward Layers Build Predictions by Promoting Concepts in the Vocabulary Space. *EMNLP 2022*.
- Heimersheim, S., and Nanda, N. (2024). How to Use and Interpret Activation Patching. arXiv:2404.15255.
- Joshi, A., Ahmad, A., and Modi, A. (2025). Calibration Across Layers: Understanding Calibration Evolution in LLMs. *EMNLP 2025*.
- Lad, V., Lee, J. H., Gurnee, W., and Tegmark, M. (2025). The Remarkable Robustness of LLMs: Stages of Inference? *NeurIPS 2025*.
- Lindsey, J., Templeton, A., Marcus, J., Conerly, T., Batson, J., and Olah, C. (2024). Sparse Cross-coders for Cross-Layer Features and Model Diffing. *Transformer Circuits Thread*.
- Minder, J., Dumas, C., Juang, C., Chughtai, B., and Nanda, N. (2025). Robustly Identifying Concepts Introduced During Chat Fine-Tuning Using Crosscoders. arXiv:2504.02922.
- Prakash, N., Shaham, T. R., Haklay, T., Belinkov, Y., and Bau, D. (2024). Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking. *ICLR 2024*.
- Wu, X., Yao, W., Chen, J., Pan, X., Wang, X., Liu, N., and Yu, D. (2024). From Language Modeling to Instruction Following: Understanding the Behavior Shift in LLMs after Instruction Tuning. *NAACL 2024*.
- Zhao, Z., Ziser, Y., and Cohen, S. B. (2024). Layer by Layer: Uncovering Where Multi-Task Learning Happens in Instruction-Tuned Large Language Models. *EMNLP 2024*.

Appendix Roadmap

Appendix	Supports	Does not prove
A. Convergence-gap robustness	raw/tuned discovery, endpoint matching, and endpoint-free same-history controls	deployment-level behavior
B. Matched-prefix MLP leverage	late graft/swap effects, random controls, and width/center robustness	full circuit recovery or a sharp late-only boundary
C. Gemma behavioral case study	free-running effect of the late PT swap on one preselected high-signal family	cross-family behavioral generalization
D. Reproducibility map	CPU claim checks and artifact hashes	hardware-free full raw reruns

A Convergence-Gap Robustness

The convergence-gap claim is carried by a compact ladder: native discovery curves, endpoint-matched late KL, endpoint-free adjacent JS, and endpoint-free future top-1 flips. Commitment thresholds are reported as summaries of the same trajectory and are not used as a single decisive definition.

The endpoint-matched control matches token steps by final confidence, entropy, and top1/top2 margin. Minimum matched retention is 0.796; maximum post-match standardized mean difference is 0.057. The remaining IT-minus-PT late gap is therefore not a simple artifact of more confident or less confident endpoints.

Template-regime fixed-history audit: instruction-tuned checkpoints are normally trained and released for a serialized chat interface, so native chat-template prompting is the main condition rather than an optional wrapper. To remove rollout-length confounding, we generated one IT-native continuation per prompt and replayed the same forced token IDs through `pt_raw`, `it_native`, and `it_raw` cells for the four dense families with completed IT-native replay records. Same-prompt/step raw-lens late-KL effects are +1.181 nats [+1.153, +1.211] for `it_native - pt_raw`, +0.549 [+0.531, +0.568] for `it_raw - pt_raw`, and +0.632 [+0.611, +0.654] for `it_native - it_raw`. Endpoint-matched CEM effects are +0.548 [+0.502, +0.594], +0.202 [+0.158, +0.247], and +0.357 [+0.306, +0.411], respectively, with 0.999 minimum retention, 0.061 maximum SMD, zero malformed records, and zero missing aligned token steps. The native-template gap therefore survives fixed-continuation replay; raw/no-template serialization remains positive but attenuated, so it is robustness and limitation evidence rather than a replacement for the trained IT interface.

As a reverse teacher-source stress test, we also generated PT-raw continuations and replayed those histories through the same three cells. The paired raw-lens late-KL effects have the same sign but are weaker: +0.610 [+0.135, +1.079] for `it_native - pt_raw`, +0.429 [+0.155, +0.639] for `it_raw - pt_raw`, and +0.181 [-0.105, +0.467] for `it_native - it_raw`. Collection quality is clean, with zero malformed records and zero missing aligned steps, but endpoint balance is not uniformly within the pre-specified SMD threshold (0.991 minimum retention, 0.155 maximum SMD). We therefore use this reverse replay only as same-sign support under PT histories, not as stronger evidence for the template-delta claim.

The fixed-history replay table is generated from the checked synthesis artifacts:

Teacher history	Estimator	IT native - PT raw	IT raw - PT raw	IT native - IT raw	Use/quality
IT-native continuation	paired same prompt/step	+1.181 [+1.153, +1.211]	+0.549 [+0.531, +0.568]	+0.632 [+0.611, +0.654]	primary fixed-history replay
IT-native continuation	endpoint-matched CEM	+0.548 [+0.502, +0.594]	+0.202 [+0.158, +0.247]	+0.357 [+0.306, +0.411]	retention 0.999; max SMD 0.061
PT-raw continuation	paired same prompt/step	+0.610 [+0.135, +1.079]	+0.429 [+0.155, +0.639]	+0.181 [-0.105, +0.467]	same-sign reverse stress test
PT-raw continuation	endpoint-matched CEM	+0.025 [-0.183, +0.233]	+0.012 [-0.040, +0.064]	-0.106 [-0.274, +0.111]	balance caveat: retention 0.991; max SMD 0.155

Discovery-curve token-step counts:

Family	PT token steps	IT token steps	Layers
Gemma 3 4B	1,273,606	810,347	34
Llama 3.1 8B	517,579	499,240	32
Qwen 3 4B	482,319	636,979	36
Mistral 7B v0.3	1,380,081	501,093	32
OLMo 2 7B	370,171	166,698	32
DeepSeek-V2-Lite	184,658	163,877	27

B Matched-Prefix MLP Leverage

Matched-prefix graft/swap experiments use overlapping early, middle, and late MLP windows. The dense-family causal pool excludes DeepSeek because MoE routing changes the interpretation of swapping dense MLP windows.

The random-control follow-up uses matched late residual-projection perturbations. The true late graft gives +0.327 final-20% KL change, while the matched random perturbation gives +0.003. This control is the main reason we interpret the late result as a specific matched-prefix update rather than generic late fragility.

Per-family headline late-window effects show that the dense mean is not driven by a single model:

Family	Late IT graft into PT host	Late PT swap into IT host	Late window
Gemma 3 4B	+0.609	-0.822	20-33
Qwen 3 4B	+0.491	-1.015	22-35
Llama 3.1 8B	+0.310	-0.291	19-31
Mistral 7B v0.3	+0.115	-0.273	19-31
OLMo 2 7B	+0.181	-0.142	19-31
Dense mean	+0.341	-0.509	-

The effect is heterogeneous in magnitude but sign-consistent for the late-window headline across the dense families. The current paper therefore treats the dense mean as a center-of-mass summary and lists per-family values for audit.

A late-window width/center audit reruns the matched-prefix experiment over six overlapping windows on the same five dense families (600 prompts, 128 forced steps). The table reports dense-family model-mean $KL(layer || own final)$ deltas; final-20% values measure downstream stabilization, while graft-window values measure the directly edited layers.

Window	Final-20 IT graft into PT	Final-20 PT swap into IT	Edited-window IT graft into PT	Edited-window PT swap into IT
Pre-late half	-0.001	-0.384	+0.069	-2.674
Late full	+0.070	-0.625	+0.365	-1.605
Late front half	+0.008	-0.352	+0.142	-1.321
Late center half	+0.022	-0.352	+0.094	-0.701
Late terminal half	+0.050	-0.443	+0.040	-0.403
Terminal quarter	+0.033	-0.347	-0.012	-0.134

This audit supports the late-window leverage claim but also narrows it. The late-full final-20% swap remains negative with a bootstrap CI excluding zero (-0.625 nats, [-1.076, -0.201]), and late subwindows preserve the swap-side sign. The late-full graft is positive but weaker at final-20% (+0.070, [-0.048, +0.189]); its within-window effect is clearer (+0.365, [+0.108, +0.629]), with late-front and late-center graft-window estimates also above zero. Because the pre-late swap control is also negative, the audit should not be read as proving a uniquely late-only boundary. It supports the more conservative claim made in the main text: among tested matched-prefix MLP windows, the late window is the strongest bidirectional handle on delayed stabilization.

C Gemma Behavioral Case Study

The Gemma behavioral case study reuses the same matched-prefix intervention family under free-running generation. It is included to show behavior-facing plausibility for the clearest convergence-

intervention family, not to make a population-level behavioral claim. Full dense-family Exp15 behavior is noisier on the PT-graft sufficiency side; the cleanest behavior-facing result is the IT-side late PT swap.

Readout	IT native	Late PT swap	Effect/preference	N
Assistant-register score	4.733 [4.631, 4.822]	3.929 [3.769, 4.080]	+0.804 [+0.649, +0.960]	225
Pairwise assistant-register	0.947 [0.916, 0.973]	0.053 [0.027, 0.084]	native preferred	225
Pairwise safety/format	0.933 [0.867, 0.987]	0.067 [0.013, 0.133]	native preferred	75
MMLU forced-choice accuracy	0.517	0.483	-0.033	60
Reasoning exact match	0.850	0.925	+0.075	40
Format compliance	0.574	0.412	-0.162	68

D Reproducibility and Artifact Map

We provide a CPU-only reviewer artifact bundle that regenerates the quoted paper numbers from committed JSON/CSV summaries. Full raw intervention reruns require multi-GPU hardware and are optional for audit.

Claim group	Checker/artifacts
Convergence gap	CPU claim checker; reporting-table generator; Exp9/Exp22 summary JSON/CSV; generated reporting tables
Matched-prefix MLP leverage	Exp11 depth-ablation metrics; Exp14 symmetric graft/swap summary; Exp19B random-control summary; Exp55 late-window width/center synthesis
Gemma behavioral case study	Exp15 Gemma behavior synthesis; generated reporting table

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